

FPGA BASED NON-INVASIVE HEART RATE MONITORING SYSTEM FOR DETECTING ABNORMALITIES IN FETAL

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Abstract— The aim of this paper is to extract the Fetal Electrocardiogram (FECG) signal from the mother's abdomen signal which is an important signal to analyze the abnormalities of the fetal heart by a physician. Fetal ECG is extracted from the mother's abdomen by using least-mean-square (LMS) adaptive filter based on the FIR filter coefficients. This FECG is used to calculate the fetal heart beat and the period of the signal based on our estimation. By using this system FECG can be taken after the pregnancy of 20 weeks. To take FECG in real time, maternal woman should wear a wide belt around her abdomen which is fitted with several ECG electrodes. The data collected from the electrodes are sent to a system and then analyzed with the help of adaptive noise canceller algorithm, which is used to separate the signals. This separation is here done based on predefined values by using the LabVIEW tool. And also the FIR filter has been designed and simulated using Xilinx ISE 12.1 and implemented in Spartan 3 Family FPGA.

Index Terms— *Fetal ECG; Least Mean Square (LMS); Adaptive Algorithm; LabVIEW.*

I.INTRODUCTION

Fetal Electrocardiogram (FECG) test is used to check the fetal health condition and abnormalities. This test provides the solution to give better treatment to the patient during pregnancy stage^[8]. FECG also provides the fetal heart beat to estimate health condition regular monitoring checkup is needed during the pregnancy period, from this checkup the condition of the fetal can be detected early and suitable treatment or proper decision can be taken on time^[5]. This paper will be helping the doctors to extract the fetal heart rate. Fetal Heart Rate detection can also be useful in other cases also like: detection of gestation age of the baby, monitor progress of labor detection of miscarriage rate for pregnancies, and monitor and evaluation of uterine contractions, etc. Electrocardiogram (ECG) is a simple noninvasive system for detecting the heart diseases^[3]. Non Invasive FECG is used to get valuable clinical information of fetal condition during pregnancy by placing electrodes on the maternal abdomen^[2]. Abdominal ECG (AECG) is always affected by power line interference, maternal ECG (MECG), Electromyogram (EMG) in case of FECG signal is affected by maternal ECG, maternal electromyogram, power line interference, baseline wander and random electronic noise, the gestational age, position of the electrodes and the

skin impedance^[1]. By using adaptive filtering techniques, these unwanted noises can be eliminated.

II. FETAL ECG

A.ECG TRACING

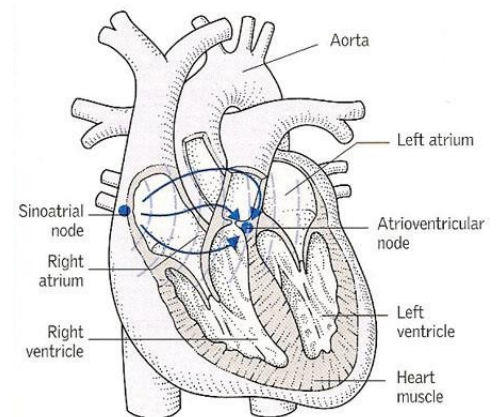


Fig 1: Human Heart Anatomy

The ECG signal provides the electrical activity of the heart. ECG signal consists of a P wave, QRS complex and T wave. The abdominal electrocardiogram signal (QRS complex) is very important signal which is used to obtain the information of the heart rate and this QRS complex is used to detect the abnormalities of the heart activities. ECG signals can be measured at two places 1). Chest and 2). Abdomen [3]. Electrocardiogram measured in the Chest contains the heart rate of the mother and the Electrocardiogram measured at the abdomen consists of composite signal of the maternal electrocardiogram (MECG) and the fetal electrocardiogram (FECG). Fetal heart rate (FHR) can be calculated by determining the R-R intervals from the fetal ECG signal^[10]. Number of heart beats per minute is higher than the mother heart beats but the amplitude of the fetal ECG is much weaker than the mother's ECG^[7]. For the extraction of FECG adaptive algorithm is used to improve the reliability. As extracted FECG is always fluctuated, R peaks cannot be detected correctly in previous works this research work overcomes the difficulty of amplitude threshold and detect the overlapped peaks^[6].

III. ADAPTIVE FILTER

An adaptive filter is a Filter which is used to separate the signals according to the adaptive algorithm. This has the advantage of user can adjust the filter co efficient as per their need.

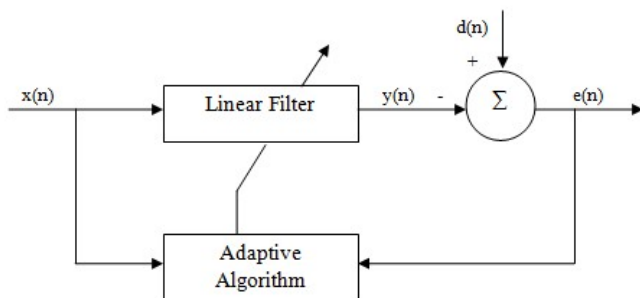


Fig 2: Diagram of a typical adaptive filter.

Where

$x(n)$ is the maternal heartbeat signal,

$y(n)$ is the corresponding output signal,

$d(n)$ is an additional input signal to the adaptive filter,

$e(n)$ is the error signal that denotes the difference between $d(n)$ and $y(n)$

IV. FIR FILTER

The Electrocardiogram is a test that is used to record the heart's electrical potential versus time. ECG signal commonly contains three types of predominant noise that are baseline wander (BW) noise, electromyography (EMG) interference, and 50 or 60 Hz power line interference. Among them power line interference is an important noise that affects the ECG and also Cables carrying ECG signals from the patients to the monitoring equipment are affected by electromagnetic interference (EMI) from the 50/60Hz power line noise. There were many methods used previously for the removal of power line interference in the ECG. They are classified into non-adaptive and adaptive filtering. The non-adaptive filtering approach contains a sharp notch filter and can be implemented at low cost. However its efficiency depends on the frequency stability of the power line. The Adaptive filtering method effectively removes time-varying power line frequency. It requires considerable computational power which is not suitable for portable battery powered ECG devices. For lower power consumption in proposed technique a non-adaptive filtering based upon a multiplication-free linear phase FIR filter is utilized. During Pregnancy stage it is a necessary checkup for the maternal and the fetal. The ECG is a simple and invasive test which is used for detecting various heart diseases. Fetal ECG signal provides the heart rate of the fetal and provides the physiological state. By placing skin electrodes on the maternal abdomen, both the fetal heart rate and maternal heart rate can be obtained. The aim is to extract the fetal heart from the maternal heart rate by using Adaptive signal processing Least Mean Square (LMS) algorithm in the

LabVIEW environment. The extracted fetal ECG from the maternal ECG by using LabVIEW tool based upon the assumption vales of the fetal and maternal heart rate. Number of heart beats per minute is higher than the mother heart beats but the amplitude of the fetal ECG is much weaker than the mother's ECG. From this study, the following commands are used to create an electrocardiogram signal of the mother heart signal assuming that the heart rate is approximately 85 beats per minute, and the peak voltage of the signal is 3.2millivolts and the sampling rate for these values will be 3900 Hz. Likewise commands are Created for the fetal also having the values of 130 beats per minute and the peak voltage of the signal is 0.25 millivolts. Here notice that fetal electrocardiogram signal of the mother's abdomen is dominated by the maternal heartbeat. This path can be explained as a 10 randomized coefficients of linear FIR filter. Small amount of uncorrelated Gaussian noise also added with this measurement to simulate the broadband noise. least-mean-square (LMS) adaptive filter with 12 coefficients and a step size of 0.00001 is used to extract the fetal ECG from the maternal abdomen ECG for this task adaptive filter needs a reference signal, the heart beat signal which is taken from the chest of the mother is act as a reference signal here. The estimated fetal heart beat signal is taken from the output signal $y(n)$ of the adaptive filter. The remaining signal which is located in the adaptive filter is the error signal $e(n)$ which contains the mother heart beat signal.

V.METHODOLOGY

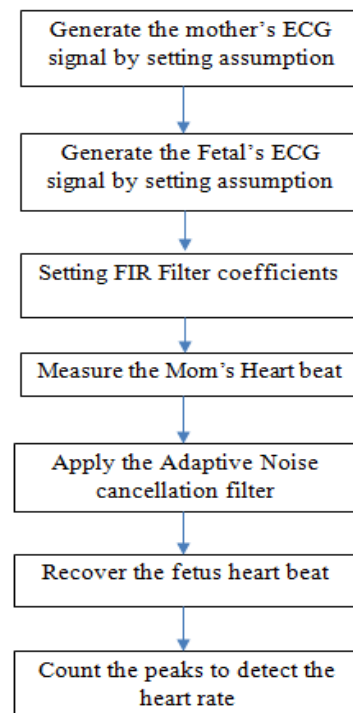


Fig 3: Flow diagram representation

Fetal ECG monitoring is a test, it is used to detect any abnormalities in the fetal heart and it gives a way to take necessary action by a physician. During Pregnancy stage it is a necessary checkup for the maternal and the fetal. The ECG is a simple and invasive test which is used for detecting various heart diseases. Fetal ECG signal provides the heart rate of the fetal and provides the physiological state. By placing skin electrodes on the maternal abdomen, both the fetal heart rate and maternal heart rate can be obtained. The aim of this is to extract the fetal heart from the maternal heart rate by using Adaptive signal processing Least Mean Square (LMS) algorithm in the LabVIEW environment. Here we have extracted fetal ECG from the maternal ECG by using LabVIEW tool based upon the assumption vales of the fetal and maternal heart rate. Number of heart beats per minute is higher than the mother heart beats but the amplitude of the fetal ECG is much weaker than the mother's ECG. From this study, the following commands are used to create an electrocardiogram signal of the mother heart signal assuming that the heart rate is approximately 85 beats per minute, and the peak voltage of the signal is 3.2 millivolts and the sampling rate for these values will be 3900 Hz. Likewise commands are created for the fetal also having the values of 130 beats per minute and the peak voltage of the signal is 0.25 millivolts. Here notice that fetal electrocardiogram signal of the mother's abdomen is dominated by the maternal heartbeat. This path can be explained as a 10 randomized coefficients of linear FIR filter. Small amount of uncorrelated Gaussian noise also added with this measurement to simulate the broadband noise. least-mean-square (LMS) adaptive filter with 12 coefficients and a step size of 0.00001 is used to extract the fetal ECG from the maternal abdomen ECG for this task adaptive filter needs a reference signal, the heart beat signal which is taken from the chest of the mother is act as a reference signal here. The estimated fetal heart beat signal is taken from the output signal $y(n)$ of the adaptive filter. The remain signal signal which is located in the adaptive filter is the error signal $e(n)$ which contains the mother heart beat signal.

VI. BLOCK DIAGRAM

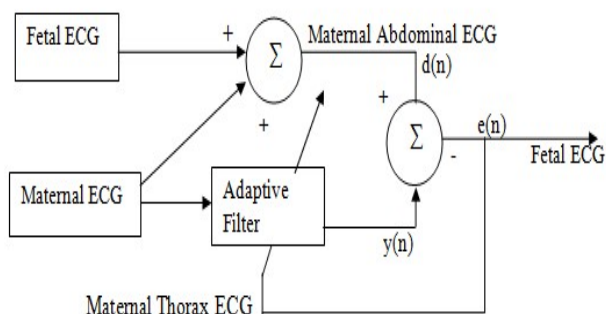


Fig 4: Cancelling the Maternal ECG from Fetal ECG

To design digital filters requires approved specification with fixed coefficients are needed when this specifications are not accessible or time varying then this problem overcomes by

adaptive coefficient this is called as adaptive filter. In modern times, Digital signal processing is used for the applications of biomedical engineering where the unwanted signal from the original ECG signal can be removed by using digital filters. But now adaptive signal processing is used to implement different algorithms. Noise cancellation is main problem in the biomedical system which is considered as adaptive noise cancellation. Adaptive filtering method can be operated by setting the parameters. The filters are important part of the system which performs any kind of manipulation or signal processing to neglect any unwanted signal or noise created in the signal. So the digital filters having same appearance such as adaptive filtering, which provide better performance by adjusting the filter coefficients.

VII. LABVIEW

LabVIEW (Laboratory Virtual Instrument Engineering Workbench) is a graphical programming language, developed by National Instruments. By using this tool we can develop applications by using icons instead of programming code. This application will be saved as an extension of .vi (Virtual Instruments). The major areas of LabVIEW include Data acquisition, signal Processing, and hardware control. LabVIEW is oftenly used for data acquisition, instrument control and industrial automation.

A.FRONT PANEL

Front panel is a user interface window, where the user or designer can interact and give the inputs in this window. After the design the user can view the output in front panel itself. It contains the icons of appropriate functions this window can be designed as per the users need.

B.BLOCK DIAGRAM

Block diagram is not a user interface window, it contains the functions of graphical code that will not be modified by the designer. This is the main window, this window replaces the traditional programming codes (like C, C++) with icons. The user interface icons are called Controls this will be in front panel window, the same icon will be act as indicator in Block Diagram.

VIII. LABVIEW RESULTS

The below figure 5 shows the front panel window of Fetal ECG extraction where the input and outputs are displayed. Here two inputs are given namely filter length of 40 and step size of 0.45 and their corresponding output ECG signals are taken. Here four different signals are displayed, in that the first graph shows the Maternal Thorax ECG signal which is the maternal heartbeat signal

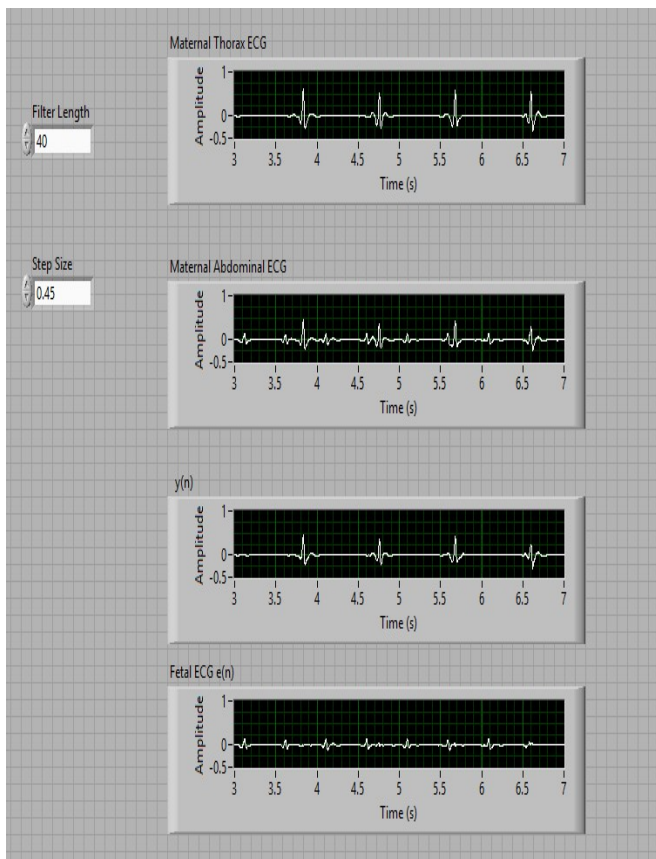


Fig 5: Front Panel window of Fetal ECG extraction using LabVIEW

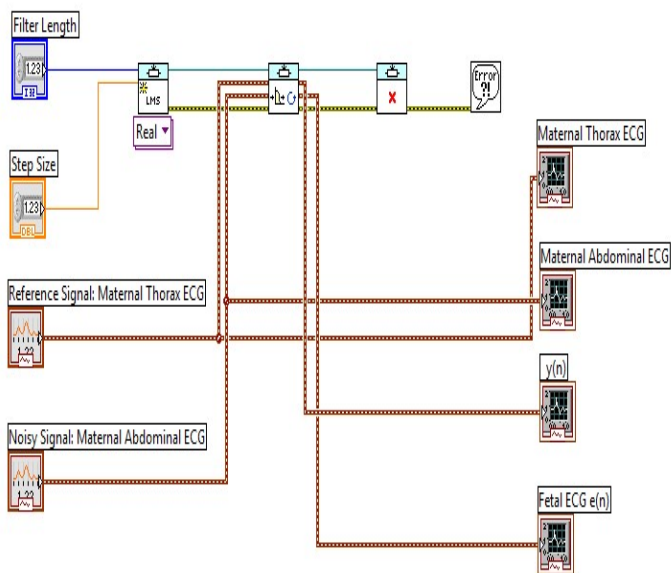


Fig.6: Block Diagram window of Fetal ECG extraction using LabVIEW

Second graph shows the Maternal abdominal ECG signal which is taken from the abdomen of the maternal woman. this signal contains the heart beat signals of both fetal and mother. The both 1st and 2nd graphs are the input signal for adaptive filter.

Third graph shows the error signal $y(n)$ which is separated by the adaptive algorithm. Here maternal heart beat signal is the error signal which is approximately equal to the 1st Graph.

Fourth graph shows our require fetal ECG signal which is separated from the 2nd graph using adaptive filter.

The above figure 6 shows the Block Diagram window of Fetal ECG which contains the functions. The icons of filter length and Step size are applied to the Adaptive LMS algorithm. The Maternal Thorax ECG and Maternal Abdominal ECG are applied to the adaptive filter and also the outputs of LMS are applied to this adaptive filter. This adaptive filter separates the signals and gives the outputs.

XI. VERILOG OUTPUT

The Figure 7 Shows the FIR filter has been designed and simulated using Xilinx ISE 12.1 and implemented in Spartan 3 Family FPGA.

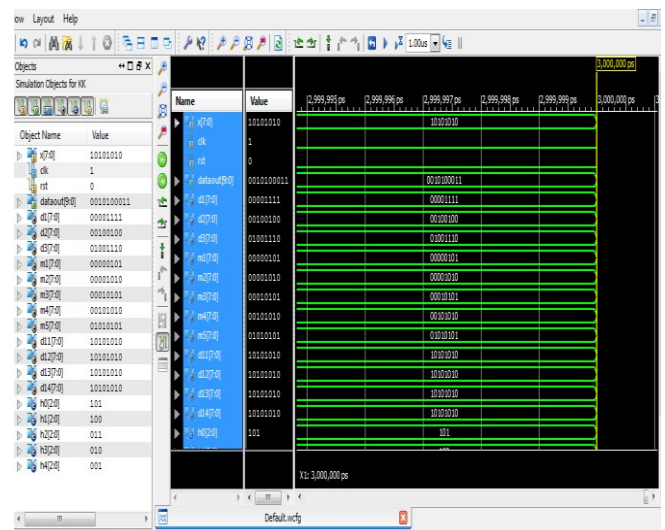


Fig 7: FIR Filter Implementation in Verilog

XII. FUTURE ENHANCEMENT

Here the mother and fetal ECG are obtained by setting predefined values in LabVIEW Platform. In future this Fetal ECG can be implemented in hardware's like MSP430 by using Skin Electrodes in Real time.

XIII.CONCLUSION

An advanced Fetal Electrocardiogram (FECG) signal monitoring for both the mother and fetal analysis system has been designed by using an LMS adaptive algorithm based on FIR filter which is used to estimate the baby's heartbeat and calculate the period of the signal. The FECG signals have been extracted from the abdominal electrocardiogram signals using LabVIEW software. The implementation in FPGA platform results low power consumption and utilized minimum hardware resources.

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